Unit 9: Interval Estimation

8.5: General (The Pivotal Method)

8.6-8.7: Large-Sample (*Z-tools*) Confidence Intervals

Applications: $\mu,\ \mu_1 - \mu_2,\ p,\ p_1 - p_2$

8.8: Small-Sample (*t-tools*) Confidence Intervals

Applications: $\mu,\ \mu_1 - \mu_2$

8.9: Confidence Intervals for Variances

Applications: $\sigma^2,\ \sigma_1^2/\sigma_2^2$

Definition: Confidence Interval

θ – target parameter. $\hat{\theta}$ – point estimator.

- An **interval estimate** for θ is a range or interval of estimates, denoted $[\hat{\theta}_L, \hat{\theta}_U]$.
- $\hat{\theta}_L$ and $\hat{\theta}_U$ are random variables representing the lower and upper limits of the interval estimate.

If, prior to sampling,

i) $P(\hat{\theta}_L \le \theta \le \hat{\theta}_U) = 1-\alpha,$ then the interval $[\hat{\theta}_L, \hat{\theta}_U]$ is called a two-sided $(1-\alpha)100\%$ confidence interval for θ.

ii) $P(\hat{\theta}_L \le \theta) = 1-\alpha,$ then we call $[\hat{\theta}_L, \infty)$ a lower one-sided $(1-\alpha)100\%$ confidence interval for θ. Or, $\hat{\theta}_L$ is called a lower confidence bound for θ.

iii) $P(\theta \le \hat{\theta}_U) = 1-\alpha,$ then we call $(\infty, \hat{\theta}_U]$ an upper one-sided $(1-\alpha)100\%$ confidence interval for θ. Or, $\hat{\theta}_U$ is called an upper confidence bound for θ.

- $1-\alpha$ is called the **confidence level** (coefficient). α is called the **error rate**.

Interpretation / Conclusions : Confidence Interval

After sampling we observe $\hat{\theta}_L$ and $\hat{\theta}_U$ (they are calculated from the sample).

Then,

i) Two-sided $(1-\alpha)100\%$ confidence interval for θ:

$\Rightarrow$ We are $(1-\alpha)100\%$ confident that θ is between $\hat{\theta}_L$ and $\hat{\theta}_U$.

ii) Lower one-sided $(1-\alpha)100\%$ confidence bound for θ:

$\Rightarrow$ We are $(1-\alpha)100\%$ confident that θ is above $\hat{\theta}_L$.

iii) Upper one-sided $(1-\alpha)100\%$ confidence bound for θ:

$\Rightarrow$ We are $(1-\alpha)100\%$ confident that θ is below $\hat{\theta}_U$.

Example 6:

The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a ***Normal*** distribution with unknown mean and standard deviation equal to 2 ounce.

a) Suppose a sample of size ***n*** = 9 resulted in a sample mean of 14.5 ounces. Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces? Answer the question by looking at confidence intervals for the true mean.

$\mu -$ target parameter. $\bar{Y} -$ point estimator.

Consider a two-sided 95% confidence interval for μ:

- Since, prior to sampling, $P\left(\bar{Y} - 1.96\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{Y} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95,$ the interval $\left(\bar{Y} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{Y} + 1.96\frac{\sigma}{\sqrt{n}}\right)$ is called a two-sided 95% confidence interval for μ.

- Based on our sample ($n = 9$ and $\bar{y} = 14.5$) we can be 95% confident that μ is between

$$\Rightarrow \left(14.5 - 1.96\frac{2}{\sqrt{9}}, 14.5 + 1.96\frac{2}{\sqrt{9}}\right) \Rightarrow (13.19, 15.81).$$

Pivotal Method

θ – target parameter

1. Find a **pivotal quantity**. A pivotal quantity is a function $U = g(Y_1, ..., Y_n, \theta)$ that has a probability distribution independent of θ.

2. Find values a and b that satisfy $P(a < U < b) = 1 - \alpha$.

3. Solve $P(a < U < b)$ for $P(\hat{\theta}_L < \theta < \hat{\theta}_U)$.

<u>In Example 6:</u> $\mu -$ target parameter where $Y \sim Normal(\mu, \sigma^2)$.

1. Find a **Pivotal quantity :**

- $U = g(Y_1, ..., Y_n, \mu) = \dfrac{\bar{Y} - \mu}{\sigma / \sqrt{n}} \sim Normal(0,1).$

2. Find values a and b that satisfy $P(a < U < b) = 0.95$.

- $P(-1.96 < U < 1.96) = 0.95.$

3. Solve $P(a < U < b)$ for $P(\hat{\mu}_L < \mu < \hat{\mu}_U)$.

- $P(a < U < b) = P\left(-1.96 < \dfrac{\bar{Y} - \mu}{\sigma / \sqrt{n}} < 1.96\right)$

$$= P\left(\bar{Y} - 1.96\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{Y} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95.$$

Example 1:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed $Exponential(\theta)$ random variables.

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{\theta} e^{-y/\theta} & , \ y > 0 \\ 0 & , \text{else} \end{cases}$$

a) Define confidence interval(s) for θ as a function of $n = 1$ observation. Find 90% confidence interval(s).

b) Define confidence interval(s) for θ as a function of $\bar{Y}$.
($Make\ use\ of\ the\ Chi\text{-}Square\ distribution.$)

c) Define confidence interval(s) for θ as a function of $Y_{(1)}$.